

考试科目名称 软件系统设计 (A 卷)

考试方式: 闭卷 考试日期 2022 年 6 月 日 教师 张贺、潘敏学

系(专业) 软件工程 年级 大三 班级

学号 姓名 成绩

题号	一	二	三	四	五	六	七	八	九	十
分数										

得分 一、 简答题 (共 60 分)

1. Please explain the Liskov Substitution Principle and how it contributes to the Open-Closed Principle. (6 points)
2. Please explain how the Factory Method Pattern and Abstract Factory Pattern adhere to the Open-Closed Principle. (6 points)
3. What is the benefit of decoupling the Receiver from the Invoker in the Command Pattern? (6 points)
4. There are two methods for propagating data to observers with the Observer design pattern: the push model and the pull model. Why would one model be preferable over the other? What are the trade-offs of each model? (6 points)
5. What is the difference between the categories of Creational Patterns and Structural Patterns? What categories do Prototype and Flyweight fall into, respectively? Explain why. (6 points)
6. Describe the difference and relationship between software requirements, quality attributes, and architecturally significant requirements (ASR). (6 points)
7. Please define 'Availability' as a quality attribute. What do MTBF and MTTR stand for? How to calculate the availability of a system (e.g., SLA) as the probability? (6 points)
8. What are the generic design strategies applied in designing software? Give a concise working example with software architecture for these strategies. (6 points)
9. Why should a software architecture be documented using different views? Give the names and purposes of four example views. (6 points)



10. What are the risks, sensitivity points, and trade-off points associated with software architecture? Give an example for each of them. (6 points)

得分

二、设计题（共 40 分）

1. Design a Multi-instance pattern. Make sure that the number of objects of a class in the system cannot exceed a given constant n . Write code to implement a Multi-instance class. (10 points)

2. Design an OA system to store the company departments and people in a tree structure. When a notice is issued, you can set up all the people of the department or some individual people who need to be notified, and the notice will be displayed in the OA of those people. Draw the design class diagram and write the code. (10 points)

3. Figure 1 below shows a distributed application with caches to improve its runtime performance. An online customer service system provides its clients access to their Bill Statement Information (BSI). The database stores BSI for clients. During a session, a client may require the BSI several times. In order to improve the performance, the BSI information for a particular client is cached. If the cache is loaded, then the BSI can be directly returned back to the client without access to the database. There are N servers clustered to serve the requests from clients. The server is load-balanced, and each time one server is scheduled for one client request. All the caches on N servers need to synchronize their states. If the client request changes the states of one cache, there are two requirements:

REQ1. The changes are committed to the database;

REQ2. All other cache states must be invalidated and they should reload the states from the database.

The current design in Figure 1 cannot meet the second requirement. Some important components are missing. Consider to apply architectural pattern(s) and tactic(s) to the design that can meet the second requirement, and complete the following questions:

(a) Identify extra component(s) that is (are) necessary for this design to meet second requirement. List each component name and its major functionalities. (5 points)

(b) Map the roles of Observer pattern (such as subject, observer in Figure 2) to whatever necessary components including your devised component(s) and the components in Figure 1. (5 points)

(c) Draw sequence diagram to illustrate the communication between your devised components and original components in Figure 1. (5 points)

(d) Assume that the servers are heterogeneous, and the protocols to access caches on different servers vary. For example, Server 1 supports Java RMI invocations to cache 1, Server 2 supports http JSP requests to cache 2, and server 3 supports SOAP requests to cache 3. It is further assumed that the cache interfaces are not consistent to each other due to the evolution of system development and legacy problems. Explain how the concept of connectors or web services can be applied to provide a generic invocation to caches? You can choose either connector or web services for discussion. (5 points)



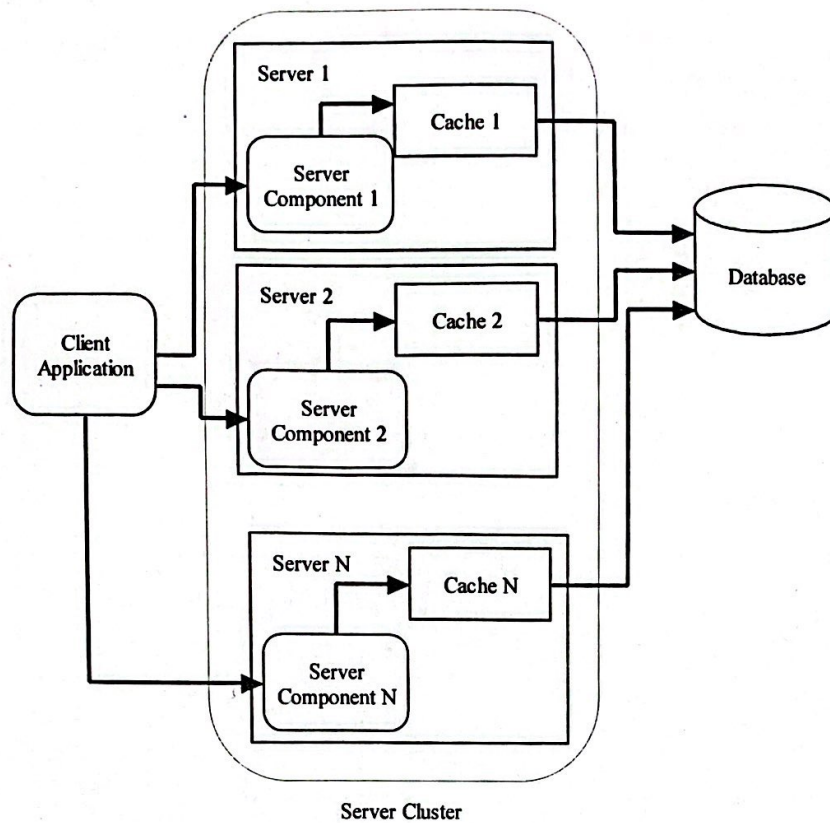


Figure 1 Distributed cache scenario

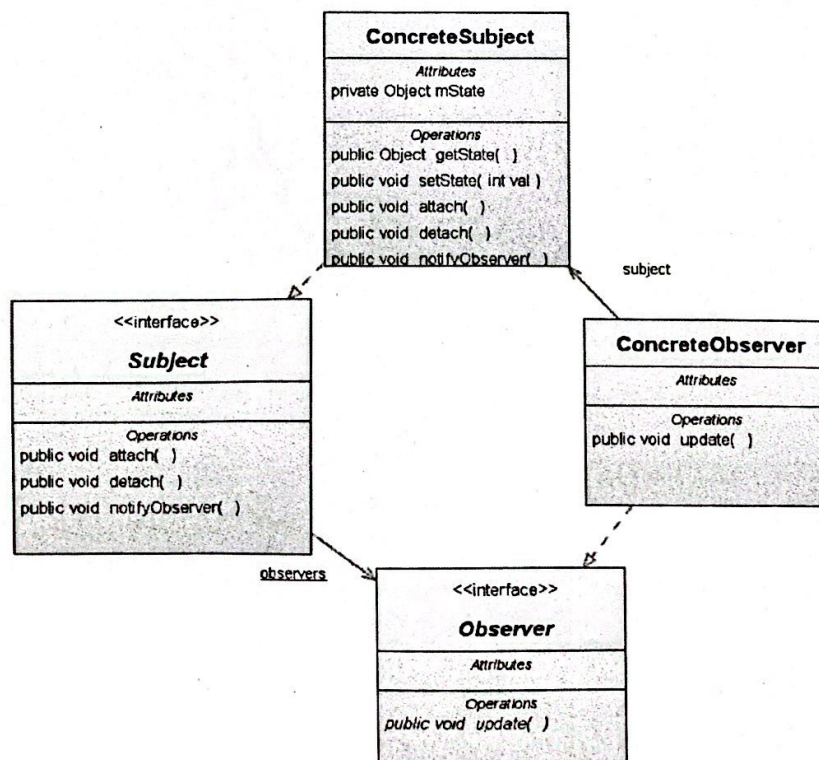


Figure 2 Observer design pattern

